
Scientific Programming LAB

Task 3: EDA Lab Task

Rodrigo, Dina, Fabian

How do they look like?



1. Load & Basic EDA - Shape, data types, summary stats, missing values

```
print(df.shape)
```

```
→ (344, 7)
```

```
print(df.head())
```

```
First 5 rows:
```

	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex
0	Adelie	Torgersen	39.1	18.7	181.0	3750.0	male
1	Adelie	Torgersen	39.5	17.4	186.0	3800.0	female
2	Adelie	Torgersen	40.3	18.0	195.0	3250.0	female
3	Adelie	Torgersen	NaN	NaN	NaN	NaN	NaN
4	Adelie	Torgersen	36.7	19.3	193.0	3450.0	female

```
print(df.info())
```

```
Data Types and Missing Values:
<class 'pandas.DataFrame'>
RangeIndex: 344 entries, 0 to 343
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  -
0   species               344 non-null   str
1   island                344 non-null   str
2   bill_length_mm        342 non-null   float64
3   bill_depth_mm         342 non-null   float64
4   flipper_length_mm     342 non-null   float64
5   body_mass_g           342 non-null   float64
6   sex                   333 non-null   str
dtypes: float64(4), str(3)
memory usage: 18.9 KB
None
```

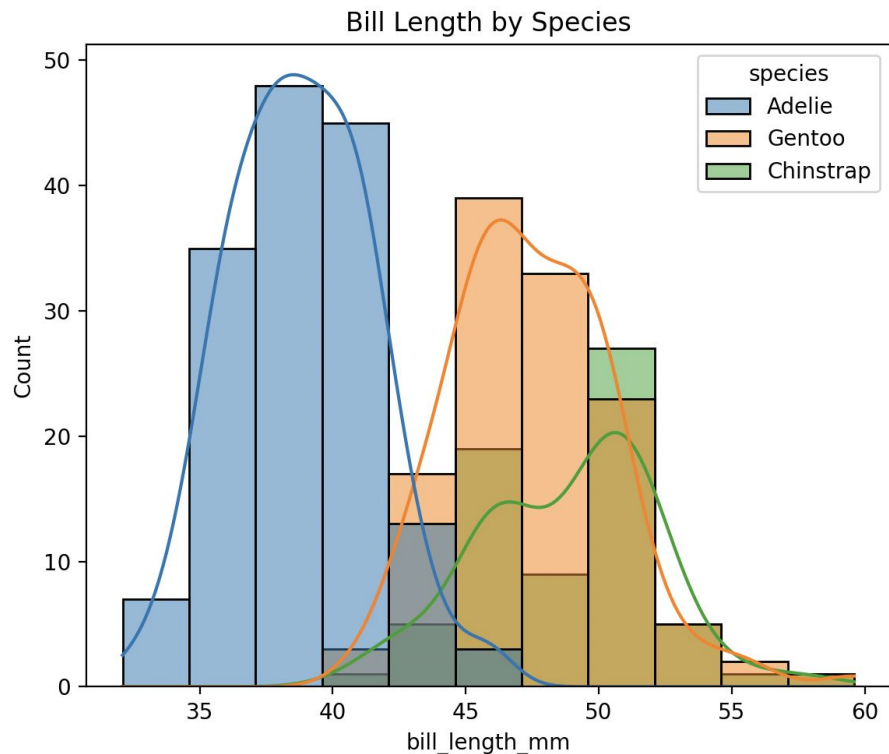
2. Data Cleaning

```
--- Missing Values Before Cleaning ---
species          0
island           0
bill_length_mm   2
bill_depth_mm    2
flipper_length_mm 2
body_mass_g      2
sex             11
dtype: int64
```

```
Missing values per column after cleaning:
species          0
island           0
bill_length_mm   0
bill_depth_mm    0
flipper_length_mm 0
body_mass_g      0
sex             0
dtype: int64
```

```
Final species counts:
species
Adelie      151
Gentoo      123
Chinstrap   68
Name: count, dtype: int64
```

Our Insights:



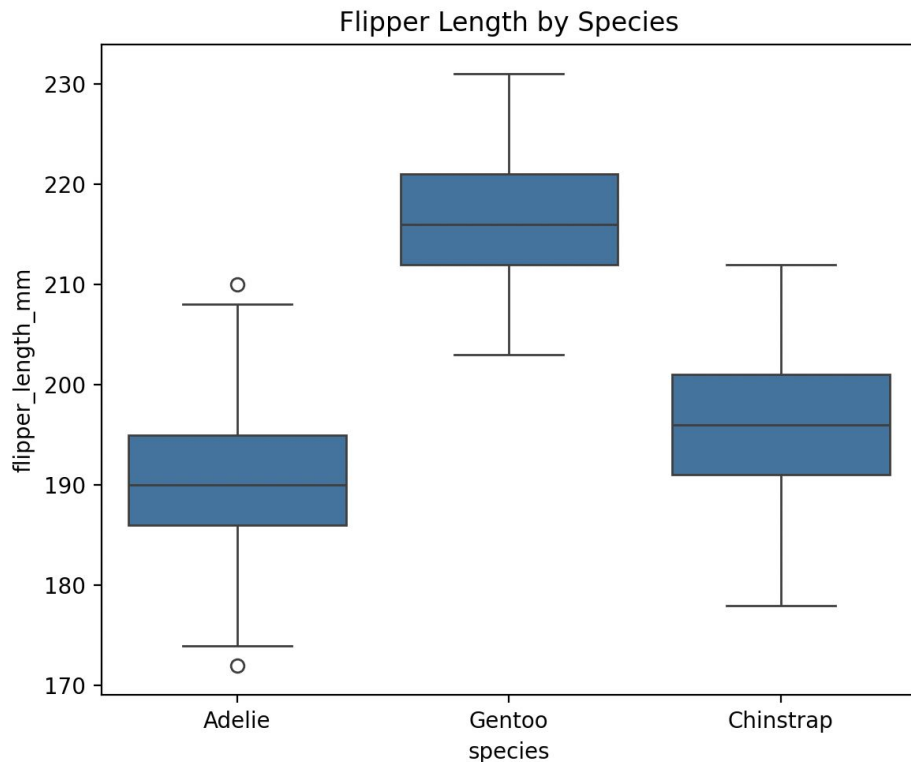
Insight:

Adelie penguins have the shortest bill lengths (mostly under 40mm), while **Gentoo** and **Chinstrap** exhibit significantly longer bills, creating a clear separation in the distribution.

Implication:

Bill length is a primary metric for identifying Adelie penguins, though secondary features are needed to distinguish between Gentoo and Chinstrap.

Our Insights:



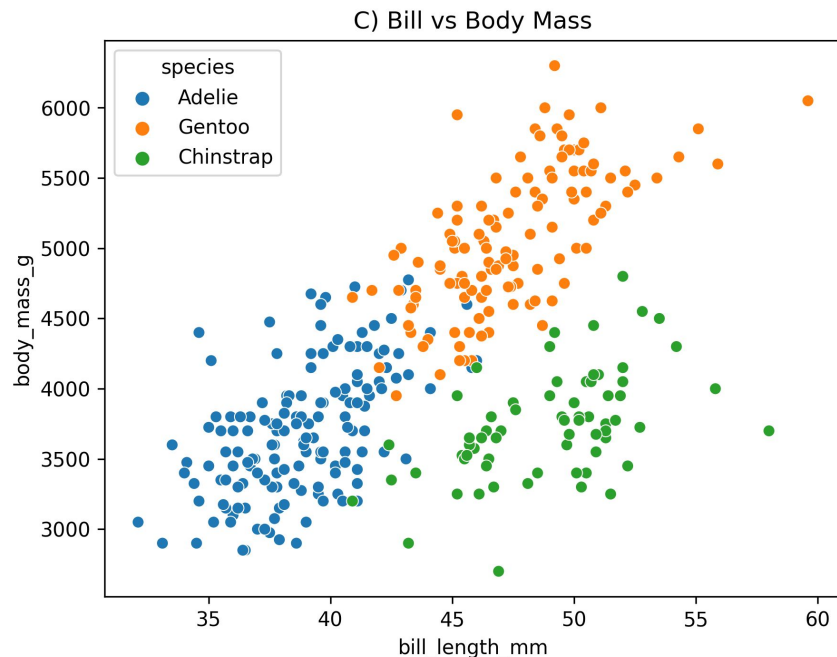
Insight:

Chinstrap penguins have intermediate flipper lengths, sitting between the smaller Adelie and the much larger Gentoo.

Implication:

Flipper length serves as a secondary distinguishing feature when bill length measurements are ambiguous.

Four Visualizations - 3



Insight:

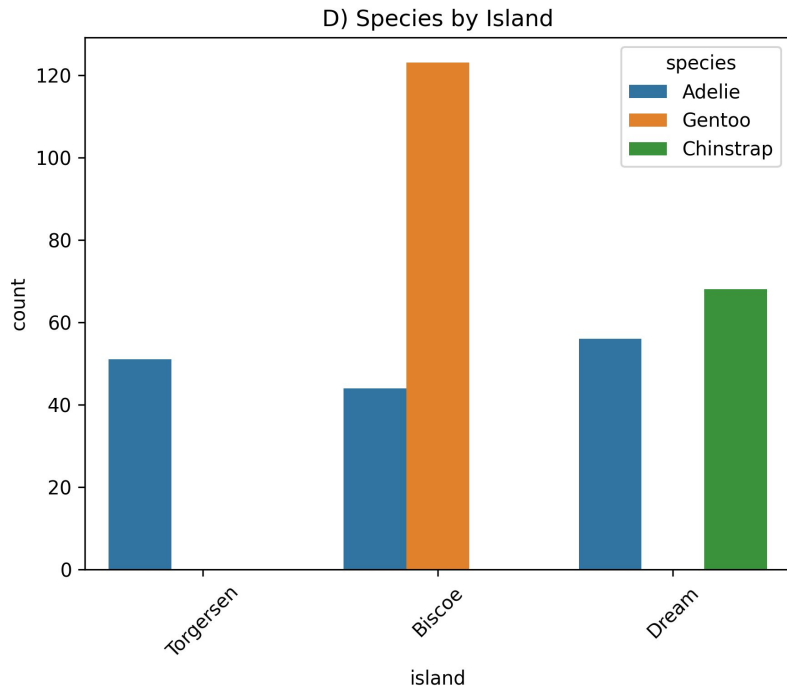
There seems to be a correlation between bill length and body mass within each species and for all species.

However, Chinstrap has longer bill length per body mass compared to Adelie and Gentoo.

Implication:

Longer mouth can eat longer fish. :)
But probably more mass $\Leftrightarrow$ taller

Four Visualizations - 4



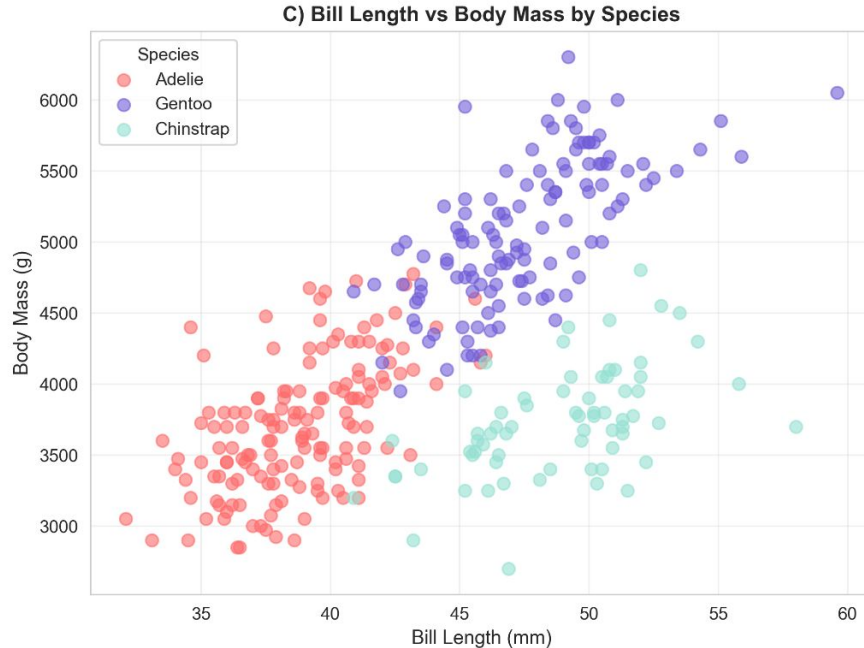
Insight:

- Torgersen: Adelie only
- Biscoe: Adelie and Gentoo, but mostly Gentoo
- Dream: Adelie and Chinstrap

Implication:

You need to go to the right place to see your favourite penguin species. They all are cute though. :)

Our Insights:



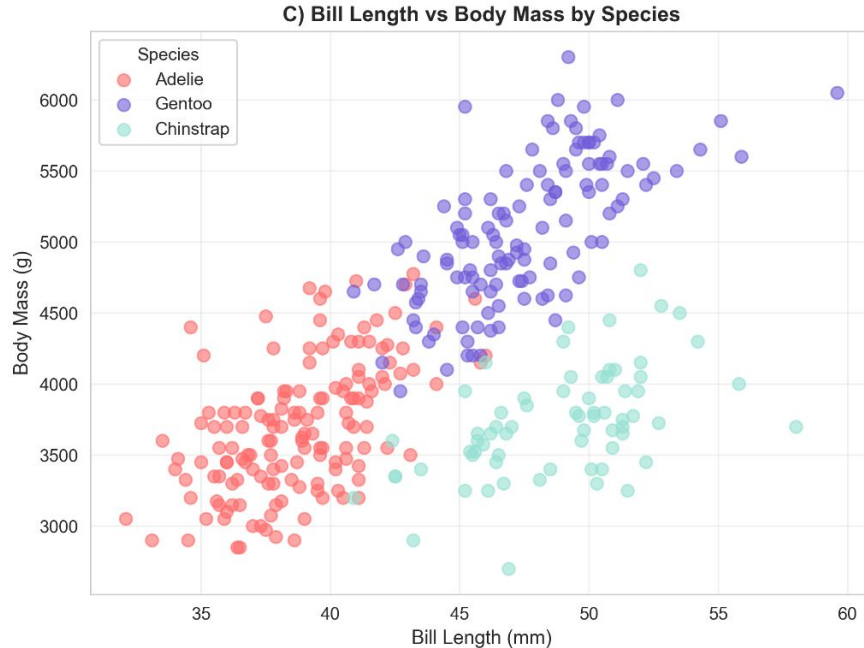
Insight:

The combination of Bill length with Body mass, or flipper length can reliably discriminate between species

Implication:

This can be used on the field to more easily identify the species of a specimen; Providing clues to researchers and increasing efficiency.

Our Insights:



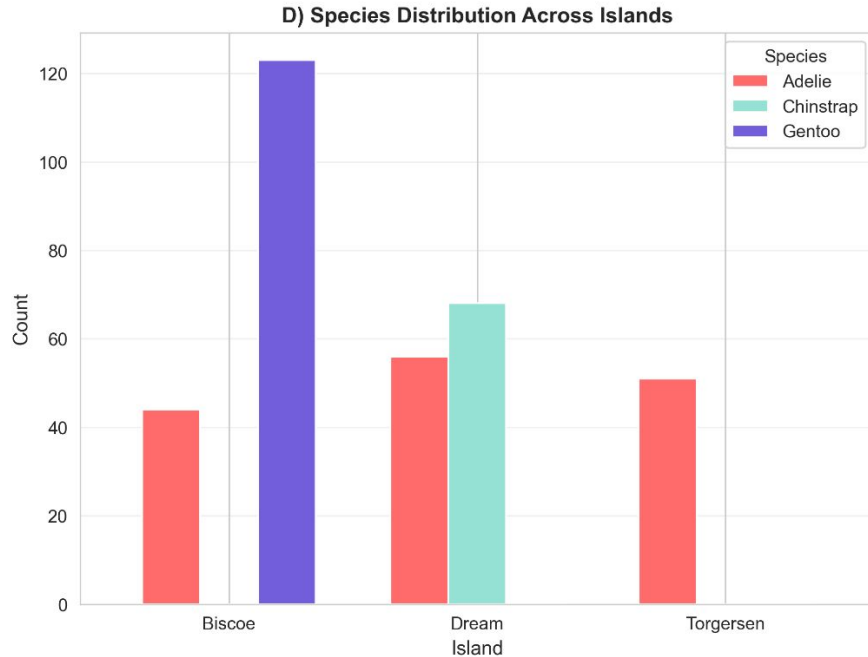
Insight:

There seems to be a strong correlation between Bill length and body mass when looking at any individual species.

Implication:

One can predict the Body mass of a specimen by knowing its species and bill length. This could potentially reduce the number of times specimens are weighed.

Our Insights:



Insight:

Adelie penguins live in all three islands, while Chinstrap and Gentoo seem to have defined territories.

Implication:

Any event which may impact the islands where Chinstrap or Gentoo penguins live could have big repercussions in the population. Adelie penguins seem to be more resilient to such events.

Conservation efforts should reflect this and implement island specific policies.